

1 Introduction

We study the interpolated stochastic heat and wave equation (ISHWE) for spatial dimension on $\mathbb{R}^d$,

$$\begin{cases} \left(\partial_t^\beta + \frac{\nu}{2}(-\Delta)^{\alpha/2} \right) u(t, x) = I_t^\gamma \left[\sigma(u(t, x)) \dot{W}(t, x) \right] & x \in \mathbb{R}^d, t > 0 \\ u(0, \cdot) = u_0 & \beta \in (0, 1] \\ u(0, \cdot) = u_0, \quad \partial_t u(0, \cdot) = v_0 & \beta \in (1, 2). \end{cases} \quad (1.1) \quad \boxed{\text{E:SPDE}}$$

2 Preliminaries

For any $t \in [0, T]$ and $x \in \mathbb{R}^d$, we write (assuming $\beta \in (1, 2)$)

$$u(t, x) = J_0(t, x) + I(t, x)$$

where

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + \frac{\partial}{\partial t}[Z(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ I(t, x) &= \int_0^t \int_{\mathbb{R}^d} Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy). \end{aligned}$$

To establish the existence and uniqueness of random field solutions to (1.1), we will need to assume Dalang's condition:

$$\int_0^t ds \int_{\mathbb{R}^d} dy |Y(s, y)|^2 < \infty, \quad \text{for all } t > 0, \quad (2.1) \quad \boxed{\text{C:Dalang}}$$

which is equivalent to

$$d < 2\alpha + \frac{\alpha}{\beta} \min\{2\gamma - 1, 0\} =: \Theta. \quad (2.2) \quad \boxed{\text{E:DalangEqu}}$$

3 1-d Case

In this section we study (1.1) for $d = 1$.

3.1 Globally Lipschitz Coefficient

In this section, we assume that the coefficient σ is globally Lipschitz and therefore satisfies the following inequality:

$$|\sigma(x)| \leq c(\sigma) + L(\sigma)|x|,$$

where $c(\sigma) = \sigma(0)$ and $L(\sigma)$ is the Lipschitz constant of σ . We assume that $L(\sigma) > 0$. We now state an existence and uniqueness result proven in [2].

alityGenLip

Theorem 3.1. *Under (2.1), the SPDE (1.1) has a unique (in the sense of versions) random field solution $\{u(t, x) : (t, x) \in (0, \infty) \times \mathbb{R}^d\}$ if the initial data are such that*

$$\hat{C}_t := \sup_{t \in [0, T], x \in \mathbb{R}^d} |J_0(t, x)| < \infty. \quad (3.1)$$

Moreover, the following statements are true:

1. $(t, x) \mapsto u(t, x)$ is $L^P(\Omega)$ -continuous for all $p \geq 2$,

2. For all even integers $p \geq 2$, all $t \geq 0$ and $x, y \in \mathbb{R}^d$,

$$\|u(t, x)\|_p^2 \leq 2J_0^2(t, x) + \left[\zeta^2 + 2\hat{C}_t^2\right] \exp(C\lambda^{2/(1-\sigma)}t). \quad (3.2)$$

1and2normYZ

Lemma 3.2. [2, Theorem 4.1 and Lemma 5.5] Assume that $d < 2\alpha$, $\beta \in (0, 2)$, and $\gamma \geq 0$. Then

$$\int_{\mathbb{R}^d} Y(t, x) dx = t^{\beta+\gamma-1}$$

and

$$\int_{\mathbb{R}^d} Y^2(t, x) dx = C_{\#} t^{2(\beta+\gamma-1)-d\beta/\alpha},$$

for all $t > 0$, where

$$C_{\#} := \frac{2}{\Gamma(d/2)(4\pi)^{d/2}} \int_0^\infty u^{d-1} E_{\beta, \beta+\gamma}^2(-u^\alpha) du.$$

Now further assume that $\beta \in (1, 2)$, then

$$\int_{\mathbb{R}} Z(t, x) dx = t^{\lceil \beta \rceil - 1} = t.$$

Remark 3.3. When $\alpha = \beta = 2$, $\gamma = 0$ and $d = 1$ then Lemma 3.2 gives us that $\int_{\mathbb{R}} G^2(t, x) dx = t/2$ which coincides with the $L^2(\mathbb{R})$ norm of the wave kernel.

In the proof of the next proposition, we will use the following two facts:

$$\sup_{t \geq 0} t^k e^{-at} = k^k (ea)^{-k} \text{ for } k \in \mathbb{N} \text{ and } \sup_{t \geq 0} \int_0^t s e^{-as} ds = a^{-2} \text{ for } a > 0. \quad (3.3)$$

E: SUPident i

I believe that Proposition 3.4 below can be proven for a more general dimension d .

Proposition 3.4. [6, Proposition 3.2] Let u_0 and v_0 be Borel functions satisfying $\|u_0\|_\infty + \|v_0\|_\infty < \infty$. Suppose that $d < 2\alpha$. Then there exists a universal constant $K := K(\alpha, \beta, \gamma) > 0$ such that for any

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))}.$$

and

$$p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2}\right].$$

we have that

$$N_{a,p}(u) \leq 2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)} \quad (3.4)$$

E: Nbdd

where

$$\mathcal{T}_0 = \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases}$$

Moreover for any $T > 0$,

$$\sup_{x \in \mathbb{R}} \mathbb{E}(|u(t, x)|^p) \leq \exp(apt) \left[2\mathcal{T}_0 + \frac{c(\sigma)}{L(\sigma)}\right]^p, \quad t \in [0, T]. \quad (3.5)$$

E: Pnormtbdd

Proof. First we fix an $a > 0$ and $p \in [2, \infty)$. Using Lemma 3.2 we see that,

$$\begin{aligned} |J(t, x)| &= \begin{cases} |(Z(t, \cdot) * u_0)(x)| & \beta \in (0, 1] \\ |(Z(t, \cdot) * v_0)(x) + \frac{\partial}{\partial t}(Z(t, \cdot) * u_0)(x)| & \beta \in (1, 2) \end{cases} \\ &\leq \begin{cases} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty & \beta \in (0, 1] \\ t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2) \end{cases} \end{aligned}$$

Now using (3.3), we see that

$$\begin{aligned} N_{a,p}(J) &\leq \begin{cases} \sup_{t \geq 0} \sup_{x \in \mathbb{R}} t^{\lceil \beta \rceil - 1} \|u_0\|_\infty e^{-at} & \beta \in (0, 1] \\ \sup_{t \geq 0} \sup_{x \in \mathbb{R}} (t^{\lceil \beta \rceil - 1} \|v_0\|_\infty + \|u_0\|_\infty) e^{-at} & \beta \in (1, 2) \end{cases} \\ &= \begin{cases} (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|u_0\|_\infty & \beta \in (0, 1] \\ (\lceil \beta \rceil - 1)^{\lceil \beta \rceil - 1} (ea)^{1 - \lceil \beta \rceil} \|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1, 2). \end{cases} \end{aligned} \quad (3.6) \quad \boxed{\text{E:Jbdd}}$$

Now we find an upper bound for $N_{a,p}(I)$. By simultaneously applying the Burkholder-Davies-Gundy inequality and triangle inequality, we get that

$$\begin{aligned} \|I(t, x)\|_p^2 &\leq 4p \left\| \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t-s, x-y) \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^t ds \int_{\mathbb{R}^d} dy Y^2(t-s, x-y) \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying Lemma 3.2 and the Lipschitz property of σ we get that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \int_0^t (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha} ds + 8pL^2(\sigma)N_{a,p}^2(u) \int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t-s, x-y).$$

Now under Dalang's condition [2], $(d/\alpha - 2)\beta - 2\gamma < -1$, the first integral above can be calculated as

$$\int_0^t (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha} ds = \frac{\alpha t^{(2-d/\alpha)\beta+2\gamma-1}}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d}.$$

By applying Lemma 3.2, we can calculate the second integral as

$$\int_0^t ds \int_{\mathbb{R}^d} dy e^{2as} Y^2(t-s, x-y) = C_\# \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha}.$$

Putting this together gives us that

$$\|I(t, x)\|_p^2 \leq 8pc^2(\sigma)C_\# \frac{\alpha t^{(2-d/\alpha)\beta+2\gamma-1}}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d} + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{2as} (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha}.$$

Now by multiplying both sides by e^{-2at} , we see that

$$\begin{aligned} \|I(t, x)\|_p^2 e^{-2at} &\leq \frac{8pc^2(\sigma)C_\# \alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d} t^{(2-d/\alpha)\beta+2\gamma-1} e^{-2at} \\ &\quad + 8pL^2(\sigma)N_{a,p}^2(u)C_\# \int_0^t ds e^{-2a(t-s)} (t-s)^{2(\beta+\gamma-1)-\beta d/\alpha}. \end{aligned}$$

We now do a change of variables and again recall the definition of the Gamma function to see that

$$\|I(t, x)\|_p^2 e^{-2at} \leq \frac{8pc^2(\sigma)C_\# \alpha}{2\alpha\beta + 2\alpha\gamma - \alpha - \beta d} t^{(2-d/\alpha)\beta+2\gamma-1} e^{-2at}$$

$$\begin{aligned}
& + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#}\int_0^\infty du e^{-2a(u)}(u)^{2(\beta+\gamma-1)-\beta d/\alpha}. \\
& = \frac{8pc^2(\sigma)C_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta d}t^{(2-d/\alpha)\beta+2\gamma-1}e^{-2at} \\
& \quad + 8pL^2(\sigma)N_{a,p}^2(u)C_{\#}(2a)^{1+(d/\alpha-2)\beta-2\gamma}\Gamma[(2-d/\alpha)\beta-1+2\gamma]
\end{aligned}$$

where the last line is true if Dalang's condition [2] holds. If we now apply (3.3) and also recall that $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, then we see that

$$\begin{aligned}
N_{a,p}(I) & \leq c(\sigma)\sqrt{\frac{8pC_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta d}}[(2-d/\alpha)\beta/2+\gamma-1/2]^{(2-d/\alpha)\beta/2+\gamma-1/2}(ea)^{1/2-\gamma-(2-d/\alpha)\beta/2} \\
& \quad + L(\sigma)N_{a,p}(u)\sqrt{8pC_{\#}(2a)^{1+(d/\alpha-2)\beta-2\gamma}\Gamma[(2-d/\alpha)\beta-1+2\gamma]}. \tag{3.7}
\end{aligned}$$

E:SoleMil3..

To simplify things, we now let

$$\begin{aligned}
K & := \max \left\{ \sqrt{\frac{8C_{\#}\alpha}{2\alpha\beta+2\alpha\gamma-\alpha-\beta d}} \left(\frac{(2-d/\alpha)\beta/2+\gamma-1/2}{e} \right)^{(2-d/\alpha)\beta/2+\gamma-1/2}, \right. \\
& \quad \left. \left(\frac{1}{2} \right)^{(2-d/\alpha)\beta/2+\gamma-1/2} \sqrt{8C_{\#}\Gamma[(2-d/\alpha)\beta-1+2\gamma]} \right\}
\end{aligned}$$

then,

$$N_{a,p}(I) \leq \frac{K\sqrt{p}}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}}(c(\sigma) + L(\sigma)N_{a,p}(u)). \tag{3.8}$$

E:Ibdd

Now by combining (3.6) and (3.8), we see that

$$N_{a,p}(u) \leq \begin{cases} (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|u_0\|_\infty & \beta \in (0,1] \\ (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1,2). \end{cases} \tag{3.9}$$

$$+ K \frac{\sqrt{p}c(\sigma)}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}} + K \frac{\sqrt{p}L(\sigma)}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}}N_{a,p}(u). \tag{3.10}$$

E:Nbddgen

It is important to note that if we now restrict ourselves to the case of $\alpha = \beta = 2$, $d = 1$ and $\gamma = 0$, then (3.7) and (3.10) recover [6, (3.9) and (3.10)] respectively. Now choose $a > 0$ such that

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))}.$$

This implies that the following interval is nonempty:

$$\left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right].$$

In addition, for any p in this interval, we have that

$$\frac{\sqrt{p}L(\sigma)K}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2},$$

and we also have

$$\frac{\sqrt{p}}{a^{(2-d/\alpha)\beta/2+\gamma-1/2}} \leq \frac{1}{2KL(\sigma)}.$$

This along with (3.10) implies that

$$N_{a,p}(u) \leq 2 \begin{cases} (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|u_0\|_\infty & \beta \in (0,1] \\ (\lceil\beta\rceil-1)^{\lceil\beta\rceil-1}(ea)^{1-\lceil\beta\rceil}\|v_0\|_\infty + \|u_0\|_\infty & \beta \in (1,2) \end{cases}$$

$$+ \frac{c(\sigma)}{L(\sigma)}$$

and this is precisely (3.4). Note that (3.5) follows immediately from (3.4). $\square$

We will need to recall the following result proved in [2] in order to prove Proposition 3.7 below.

P:YInc

Proposition 3.5. [1, Proposition 2.1 and 2.2] *Suppose that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, and (2.2) holds. Then $Y(t, x) = Y_{\alpha, \beta, \gamma, d}(t, x)$ satisfies the following:*

P:YIncCase1

1. *Suppose that $0 < \theta < (\Theta - 1) \wedge 2$ where Θ is given in (2.2), $0 < t, r \leq T$ for some $T > 0$, $\beta \leq 1$ and $\gamma \leq \lceil \beta \rceil - \beta$. Then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^\rho + |x-z|^\theta \right),$$

where

$$\rho := 2(\beta + \gamma) - 1 - \beta d / \alpha. \quad (3.11)$$

D:rho

P:YIncCase2

2. *If we only assume that $\alpha \in (0, 2]$, $\beta \in (0, 2)$, $\gamma > 0$, $0 < t, r \leq T$ for some $T > 0$ and (2.2) holds then there exists some $C := C(\alpha, \beta, \gamma, \nu, d, \theta, T)$ such that*

$$\int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \leq C \left(|t-r|^q + |x-z|^\theta \right),$$

where

$$0 < q < \rho.$$

Remark 3.6. In Proposition 3.5 above, we may often ignore the dependence of C on $\alpha, \beta, \gamma, \nu, d$ and just write $C(\theta, T)$.

P:NormInc

Proposition 3.7. *Suppose that $u_0 = 1$ and that $v_0 = 0$ so that $u(t, x) = 1 + I(t, x)$. Let $a > 0$, $\rho = 2(\beta + \gamma) - 1 - \beta d / \alpha$ and suppose that $0 < \theta < (\Theta - 1) \wedge 2$. Then for any $x, z \in \mathbb{R}$, $p \geq 2$ and for any $T > 0$ with $t \in [0, T]$,*

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^q + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)], \quad (3.12)$$

E:NormInc_N

where

$$\mathcal{M}_1 = \sqrt{p}c(\sigma) \quad \text{and} \quad \mathcal{M}_2 = \sqrt{pL(\sigma)},$$

and

1. $0 < q \leq \rho$ under Case 1 of Proposition 3.5,
2. $0 < q < \rho$ under Case 2 of Proposition 3.5.

Moreover, for any

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))}.$$

and

$$p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right]$$

we have that

$$\frac{\|u(t, x) - u(r, z)\|_p}{(|t-r|^\rho + |x-z|^\theta)^{1/2}} \leq C(p, \theta, T) \left[\mathcal{M}_1 + \mathcal{M}_2 e^{aT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right) \right]. \quad (3.13)$$

E:NormInc_c

Proof. Note that because of the initial conditions, we see that $\|u(t, x) - u(r, z)\|_p = \|I(t, x) - I(r, z)\|_p$. We now apply Burkholder-Davies-Gundy inequality along with the triangle inequality, as was done in the proof of Proposition 3.4, to see that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 4p \left\| \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \sigma^2(u(s, y)) \right\|_{p/2} \\ &\leq 4p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \|\sigma^2(u(s, y))\|_{p/2}. \end{aligned}$$

Now by applying the Lipchitz condition on σ gives us that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \left(c^2(\sigma) + L^2(\sigma) \|u(s, y)\|_p^2 \right) \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \exp(2as) N_{a,p}^2(u) \right] \\ &\leq 8p \left[c^2(\sigma) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) \int_0^T ds \int_{\mathbb{R}^d} dy |Y(t-s, x-y) - Y(r-s, z-y)|^2 \right]. \end{aligned}$$

We now apply Proposition 3.5 to see that

$$\begin{aligned} \|I(t, x) - I(r, z)\|_p^2 &\leq 8p \left[c^2(\sigma) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right. \\ &\quad \left. + L^2(\sigma) \exp(2aT) N_{a,p}^2(u) C(\theta, T) \left(|t-r|^q + |x-z|^\theta \right) \right], \end{aligned}$$

where $C(\theta, T)$ is as in Proposition 3.5. Now by taking the square root of both sides and applying the identity $\sqrt{x+y} \leq \sqrt{x} + \sqrt{y}$ yields

$$\|I(t, x) - I(r, z)\|_p \leq 2\sqrt{2C(\theta, T)p} \left[c(\sigma) + L(\sigma) \exp(aT) N_{a,p}(u) \right] \left(|t-r|^q + |x-z|^\theta \right)^{1/2}.$$

This proves (3.12). It is clear that (3.13) follows directly from (3.12) and Proposition 3.4. $\square$

The holder continuity in the next proposition needs to be checked

P:Holder

Proposition 3.8. Suppose that u_0 and v_0 are as on Proposition 3.7 above. Then u has a version, still denoted by u , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$ where

1. $0 < q \leq \rho$ under Case 1 of Proposition 3.5,
2. $0 < q < \rho$ under Case 2 of Proposition 3.5.

Moreover, if we consider

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))} \quad \text{and} \quad p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right],$$

where K is the constant from Proposition 3.4, then we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

Proof. Proposition 3.7 implies that

$$\|u(t, x) - u(r, z)\|_p \leq C(p, \theta, T) [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)] \left(|t - r|^q + |x - z|^\theta \right)^{1/2}. \quad (3.14)$$

By Kolmogorov's continuity theorem, there exists a version of u , which we denote by u , that is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

Next, note that by triangle inequality,

$$\begin{aligned} |u(t, x)|^p &\leq 2^{p-1} (\|u_0\|_\infty^p + |u(t, x) - u_0(x)|^p) \\ &\leq 2^{p-1} \left(\|u_0\|_\infty^p + \sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - u_0(x)|^p) \right) \\ &= 2^{p-1} \left(1 + \sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - 1|^p) \right) \end{aligned}$$

Due to the continuity of u on the compact set $[0, T] \times [-R, R]$, there exists a point $(t_0, x_0) \in [0, T] \times [-R, R]$ such that

$$\sup_{t \in [0, T], x \in [-R, R]} (|u(t, x) - 1|^p) = |u(t_0, x_0) - 1|^p.$$

Thus, for any $(t, x) \in [0, T] \times [-R, R]$,

$$|u(t, x)|^p \leq 2^{p-1} (1 + |u(t_0, x_0) - 1|^p)$$

which in return implies that

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \leq 2^{p-1} (1 + |u(t_0, x_0) - 1|^p).$$

Now by taking the expectation of both sides and applying Proposition 3.7 gives us

$$\begin{aligned} \mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] &\leq 2^{p-1} + 2^{p-1} C(p, \theta, T)^p \left((2T)^{\rho/2} + (2R)^{\theta/2} \right)^p [\mathcal{M}_1 + \mathcal{M}_2 e^{aT} N_{a,p}(u)]^p \\ &= 2^{p-1} + C(p, \theta, T, R) [\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} N_{a,p}(u)^p]. \end{aligned}$$

Now if we consider

$$a > (8L^2(\sigma)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))} \quad \text{and} \quad p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma)K^2} \right],$$

then by Proposition 3.4, we have that

$$\mathbb{E} \left[\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)|^p \right] \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p e^{apT} \left(2 + \frac{c(\sigma)}{L(\sigma)} \right)^p \right].$$

□

Remark 3.9. The Hölder continuity in time proven above in Proposition 3.8 recovers the result proven by Foondun and Nane in [3, Theorem 1.9] if we restrict ourself to the case where $\beta \in (0, 1)$ and $\gamma = 1 - \beta$.

Nick's Question 3.1
Then

Suppose we consider bounded initial conditions u_0 and v_0 .

$$\begin{aligned} J_0(t, x) &= \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + \frac{\partial}{\partial t}[Z(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases} \\ &=: \begin{cases} [Z(t, \cdot) * u_0](x) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) + [Z^*(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases}. \end{aligned}$$

We notice that

$$J_0(t, x) - J_0(s, z) = \begin{cases} [Z(t, \cdot) * u_0](x) - [Z(s, \cdot) * u_0](z) & \beta \in (0, 1] \\ [Z(t, \cdot) * v_0](x) - [Z(s, \cdot) * v_0](z) \\ \quad + [Z^*(s, \cdot) * u_0](z) - [Z^*(t, \cdot) * u_0](x) & \beta \in (1, 2) \end{cases}.$$

Lets just examine the $\beta \in (1, 2)$ case since that case deals with both Z and Z^* . We have that

$$|J_0(t, x) - J_0(s, z)| \leq \int_{\mathbb{R}} |Z(t, x - y) - Z(s, z - y)| |v_0(y)| dy + \int_{\mathbb{R}} |Z^*(t, x - y) - Z^*(s, z - y)| |u_0(y)| dy.$$

For simplicity lets look at the time and space increments separately. For time we have that

$$|J_0(t, x) - J_0(s, x)| \leq \int_{\mathbb{R}} |Z(t, x - y) - Z(s, x - y)| |v_0(y)| dy + \int_{\mathbb{R}} |Z^*(t, x - y) - Z^*(s, x - y)| |u_0(y)| dy.$$

Foondun and Nane [3, Proof of Theorem 1.13] stop right and claim that the above integrals are smooth functions in t . Then it seems that they conclude that they are η -Hölder continuous in t with $\eta = \frac{(1-\beta\gamma/\alpha)}{2}$ where γ is the Reisz Kernel exponent. In other words they say that

$$|J_0(t, x) - J_0(s, x)| \leq C|t - s|^\eta.$$

Are smooth functions also η -Hölder continuous for any $\eta > 0$? If we can argue as in [3], then we can relax our initial conditions to be bounded measurable functions u_0 and v_0 . This is also an improvement from Sanz-Sole and Millet [6] since they assume the initial conditions are continuous.

3.2 Existence and Uniqueness of a Global Solution with Superlinear Diffusion Coefficient

We now impose the following condition on σ : As $|x|, |z| \rightarrow \infty$,

$$|\sigma(x) - \sigma(z)| \leq \sigma_2 |x - z| [\ln_+(|x - z|)]^\delta, \quad (3.15)$$

where $\sigma_2, \delta > 0$ and $\ln_+(z) = \ln(z \vee e)$ for $z > 0$. Further, this implies that as $|x| \rightarrow \infty$,

$$|\sigma(x)| \leq \sigma_1 + \sigma_2 |x| [\ln(|x|)]^\delta, \quad (3.16)$$

where $\sigma_1 = c(\sigma) = \sigma(0)$.

Remark 3.10. Because of the absence of the drift coefficient in (1.1), we may drop [6, Assumption C1].

Theorem 3.11. Consider the initial conditions $u_0 = 1$ and $v_0 = 0$ and suppose that σ satisfies (3.15). Then for any $M > 0$, there exists a random field solution to (1.1) in $[-M, M]$ denoted as $(u(t, x) : (t, x) \in [0, T] \times [-M, M])$. This solution is unique and satisfies

$$\sup_{t \in [0, T], x \in [-R, R]} |u(t, x)| < \infty, a.s. \quad (3.17)$$

Proof. We first solve a *truncated* version of (1.1). By this, we mean (1.1) with σ replaced by σ_N where

$$\sigma_N(x) = \sigma(x) \mathbf{1}_{\{|x| \leq N\}} + \sigma(N) \mathbf{1}_{\{x > N\}} + \sigma(-N) \mathbf{1}_{\{x \leq -N\}}.$$

Note that since σ satisfies (3.15), then σ_N is Lipschitz continuous with Lipschitz constant $L(\sigma_N) := \sigma_2 \ln(2N)^\delta$. In other words,

$$|\sigma_N(x)| \leq \sigma_1 + \sigma_2 \ln(2N)^\delta |x|.$$

Hence we can apply Theorem 3.1 and see that there exists a unique solution of the truncated version of (1.1), which we denote by $u_N := \{u_N(t, x) : (t, x) \in [0, T] \times \mathbb{R}\}$. In addition, Proposition 3.8 implies that the solution has a version, still denoted by u_N , which is η_1 -Hölder continuous in time and η_2 -Hölder continuous in space with $0 < \eta_1 < q/2$ and $0 < \eta_2 < \theta/2$.

We will now apply Proposition 3.8 to find an upper bound on the p -norm of u_N . For this, consider

$$a > (8L^2(\sigma_N)K^2)^{\alpha/(\beta(2\alpha-d)+\alpha(2\gamma-1))} \quad \text{and} \quad p \in \left[2, \frac{a^{(\beta(2\alpha-d)+\alpha(2\gamma-1))/\alpha}}{4L^2(\sigma_N)K^2}\right].$$

Then Proposition 3.8 implies that

$$\mathbb{E} \left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p \right) \leq 2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2^p(N) \exp(apT) \left(2 + \frac{\sigma_1}{L(\sigma_N)} \right) \right] \quad (3.18)$$

where we recall that

$$\mathcal{M}_1 = \sqrt{p}\sigma_1 \quad \text{and} \quad \mathcal{M}_2(N) = \sqrt{pL(\sigma_N)} = \sqrt{p\sigma_2 \ln(2N)^\delta}. \quad (3.19)$$

We now use the above to prove the existence and uniqueness of (1.1) with superlinear σ satisfying (3.15). For any $T > 0$ and $N \geq 2$ with $N \in \mathbb{N}$, we define the following stopping time:

$$\tau_N := \inf \left\{ t > 0 : \sup_{|x| \leq R} |u_N(t, x)| \geq N \right\} \wedge T. \quad (3.20)$$

The uniqueness of the solution and the [local property of the stochastic integral](#) imply that $u_N(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. Hence, $(\tau_N)_{N \geq 2}$ is increasing and bounded above by T .

We now momentarily assume that $\sup_N \tau_N = T$ which would imply that $\{t < \tau_N\} \uparrow \Omega$. On each $\{t < \tau_N\}$, define $(u(t, x) : (t, x) \in [0, T] \times \mathbb{R})$ by $u(t, x) = u_N(t, x)$ and hence $u(t, x) = u_{N+k}(t, x)$ for any $k \in \mathbb{N}$. This implies that on $\{t < \tau_N\}$

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma_N(u(s, y)) W(ds, dy).$$

However, on $\{t < \tau_N\}$, we have that $\sigma_N(u_N(t, x)) = \sigma(u(t, x))$ and so u satisfies

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad \{t < \tau_N\}.$$

Lastly, since $\{t < \tau_N\} \uparrow \Omega$, we conclude that

$$u(t, x) = 1 + \int_0^t ds \int_{\mathbb{R}} dy Y(t-s, x-y) \sigma(u(s, y)) W(ds, dy), \quad (t, x) \in [0, T] \times \mathbb{R}. \quad (3.21)$$

E:SupLinMil

It is mentioned that the stochastic integrals are no longer defined in $L^2(\Omega)$. The extension of the integral is found in Dalang and Sanz-Sole's textbook that is not yet published.

We now focus on proving that $\sup_N \tau_N = T$, or equivalently that $\mathbb{P}(\tau_N < T) \rightarrow 0$ as $N \rightarrow \infty$. To do this, note that by Chebychev's inequality

$$\begin{aligned} \mathbb{P}(\tau_N < T) &\leq \mathbb{P}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)| \geq N\right) \\ &\leq N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p\right). \end{aligned}$$

An application of (3.18) gives

$$\begin{aligned} N^{-p} \mathbb{E}\left(\sup_{t \in [0, T], x \in [-R, R]} |u_N(t, x)|^p\right) &\leq N^{-p} \left(2^{p-1} + C(p, \theta, T, R) \left[\mathcal{M}_1^p + \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p\right]\right) \\ &= \frac{2^{p-1} + C(p, \theta, T, R) \mathcal{M}_1^p}{N^p} + \frac{C(p, T, R) \mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p}. \end{aligned}$$

The first term clearly converges to 0 as $N \rightarrow \infty$. As for the second term, we expand using (3.19) to see that

$$\begin{aligned} \frac{\mathcal{M}_2(N)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{L(\sigma_N)}\right)^p}{N^p} &= \frac{\left(\sqrt{p\sigma_2 \ln(2N)^\delta}\right)^p \exp\{apT\} \left(2 + \frac{\sigma_1}{\sqrt{p\sigma_2 \ln(2N)^\delta}}\right)^p}{N^p} \\ &\sim \frac{\ln(2N)^{p\delta/2}}{N^p} \rightarrow \infty, \end{aligned}$$

where above we use the symbol $\sim$ to mean asymptotically equivalent as $N \rightarrow \infty$. This completes the proof of the Theorem. $\square$

Remark 3.12. A potential project would be to show that the solution to (1.1) with superlinear coefficient is Holder continuous. This requires to one to extend the Kolmogorov theorem to the type of stochastic integral used to define the mild solution in (3.21). However these stochastic integrals are no longer defined in $L^2(\Omega)$. Maybe a completely new theory needs to be developed to show continuity. [4, Theorem 5.5.5] may be helpful?

4 Spatially Homogeneous Noise Driven SPDE

We will now look at (1.1) where the noise is now white in time and colored in space. Informally this means that

$$\mathbb{E}\left[\dot{F}(t, x) \dot{F}(s, y)\right] = \delta_0(t-s) f(x-y)$$

where f is the Riesz kernel with parameter $\lambda \in (0, d)$:

$$f(x) := \frac{1}{|x|^\lambda} \quad \text{and} \quad \mathcal{F}f(\xi) = \frac{K_f}{|\xi|^{d-\lambda}}.$$

To prove existence and uniqueness of the solution, we will need the following lemma. The following lemma is proven by Foondun and Nane [3] in the case that

$$\beta \in (0, 1) \quad \text{and} \quad \gamma = 1 - \beta \quad (4.1)$$

A:Y=GAssump

Their proof relies on a probabilistic representation of the Green's function for the solution of (1.1). Our proof takes a different approach and relies on the asymptotic behavior of the Euler-Mittag function given in [5, Theorem 1.6]. One should note that under the assumptions (4.1) that $Y(t, x) = Z(t, x)$. However, in general, this is not the case.

Lemma 4.1. *Suppose that $\lambda < \alpha$, then there exists a constant $C > 0$ such that for any $x, y \in \mathbb{R}^d$ we have that*

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) \, dz dw \leq C \left(\frac{\nu}{2} \right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha}. \quad (4.2) \quad \boxed{\text{E:YYfBdd}}$$

Proof. Through a change of variables, we may write the left side of (4.2) as the following:

$$\int_{\mathbb{R}^{2d}} Y(t, x - z) Y(t, y - w) f(w - z) \, dz dw = \int_{\mathbb{R}^{2d}} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) \, du,$$

where $*$ denotes the spatial convolution. Now an application of Parseval's identity gives us that

$$\int_{\mathbb{R}^{2d}} Y(t, u - (x - y)) (Y(t, \cdot) * f)(u) \, du = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^{2d}} \mathcal{F}Y(t, \cdot - (x - y))(\xi) \mathcal{F}(Y(t, \cdot) * f)(\xi) \, d\xi.$$

Applying the translation property of the Fourier transform and [2, Theorem 4.1] gives us that

$$\begin{aligned} & \frac{1}{(2\pi)^d} \int_{\mathbb{R}^{2d}} \mathcal{F}Y(t, \cdot - (x - y))(\xi) \mathcal{F}(Y(t, \cdot) * f)(\xi) \, d\xi \\ &= \frac{K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^{2d}} \exp(-i(x - y)\xi) \frac{E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2}{|\xi|^{d-\lambda}} \, d\xi \\ &\leq \frac{K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^{2d}} \frac{E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2}{|\xi|^{d-\lambda}} \, d\xi \end{aligned}$$

where the last inequality follows from the non-negativity of the Euler-Mittag function. Now, by applying [5, Theorem 1.6], we see that there exists a $C > 0$ such that

$$\frac{K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^{2d}} \frac{E_{\beta, \beta+\gamma} \left(-\frac{\nu t^\beta |\xi|^\alpha}{2} \right)^2}{|\xi|^{d-\lambda}} \, d\xi \leq \frac{C K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^{2d}} \frac{1}{(1 + \frac{\nu}{2} t^\beta |\xi|^\alpha) |\xi|^{d-\lambda}} \, d\xi.$$

Another change of variables implies that

$$\frac{C K_f t^{2(\beta+\gamma-1)}}{(2\pi)^d} \int_{\mathbb{R}^{2d}} \frac{1}{(1 + \frac{\nu}{2} t^\beta |\xi|^\alpha) |\xi|^{d-\lambda}} \, d\xi \leq \frac{C K_f}{(2\pi)^d} \left(\frac{\nu}{2} \right)^{-\lambda/\alpha} t^{2(\beta+\gamma-1)-\beta\lambda/\alpha} \int_{\mathbb{R}^{2d}} \frac{1}{(1 + |\xi|^\alpha) |\xi|^{d-\lambda}} \, d\xi.$$

We see that the above integral is finite if $\lambda < \alpha$ and when this is the case, we have (4.2). $\square$

We can now prove the existence and uniqueness of the solution to (1.1) driven by the noise $\dot{F}$.

References

ChenHu21

- [1] Chen, L and Hu, G. (2021). Hölder regularity of the nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *arXiv:2105.00891*.

CHN19

- [2] Chen, L., Hu, Y., and Nualart, D. (2019). Nonlinear stochastic time-fractional slow and fast diffusion equations on $\mathbb{R}^d$. *Stochastic Process. Appl.* **129**, 5073–5112

- FN2016

[3] Foondun, M and Nane E. (2016) *Asymptotic properties of some space-time fractional stochastic equations*. Mathematische Zeitschrift 2015, **287**, 493 – 519
- Kuo

[4] Kuo, H. (2000) *Introduction to Stochastic Integration* Springer, Universitext
- Podlubny99

[5] Podlubny, I. (1999). *Fractional Differential Equations*. Mathematics in Science and Engineering, Academic Press, Vol198 (1999).
- eMillet2021

[6] Sanz-Sole, M. and Millet, A. (2021) *Global solutions to stochastic wave equations with superlinear coefficients*. Stochastic Processes and their Applications, Elsevier, 2021, 139, pp.175-211.